

MANDUKHAI ALIMAA

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EDUCATION

University of Wisconsin, Madison

Masters of Science in Computer Science

Aug 2024 – May 2026

Madison, WI

Saint Mary's University Of Minnesota

Bachelor of Arts in Computer Science and Data Analytics

Aug 2018 – May 2022

Winona, MN

SOFTWARE ENGINEERING EXPERIENCE

OpenLambda | Intern

Feb 2025 - Current

Technologies: Go, Python, Linux, Docker, GCP, CI/CD, Apache Kafka, YAML, Git

Madison, WI

- Adapted and deployed OpenLambda in a local, non-cloud environment by configuring **Linux containers**, **filesystem paths**, and **networking**, enabling isolated development and testing of serverless workloads
- Engineered a per-lambda configuration system using **YAML** and **Go**, enabling each function to define its own trigger type (e.g., HTTP, Cron, Kafka), and improving execution flexibility
- Refactored configuration loading logic to support dynamic, per-lambda overrides without modifying global state, enhancing system scalability and modularity using **Go** and **YAML parsing**
- Built a trigger registry service in **Go** to scan registered lambdas and initialize their corresponding **Cron** or **Kafka triggers**, streamlining event-driven execution across workers
- Efficiently managed the relationship between **boss** and **worker** nodes, supporting reliable **lambda distribution**, **trigger activation**, and **lifecycle coordination** across components

rtNOW | Full Stack Developer

May 2022 - Aug 2024

Technologies: Vue.js, PHP, CodeIgniter 4, SQL, JavaScript, MySQL, REST APIs, WebSockets, Git

Minneapolis, MN

- Developed and maintained production-grade telehealth applications using **PHP**, **Vue.js**, and **MySQL**, improving patient scheduling efficiency and provider communication through robust backend logic and dynamic UI rendering
- Designed and implemented modular **REST APIs** using **CodeIgniter 4**, reducing backend complexity and increasing development velocity for new healthcare-focused features
- Replaced inefficient polling with a **WebSocket**-based real-time update system, reducing server load by 80% and significantly improving data freshness and responsiveness for appointment management
- Contributed to user-facing components using **Vue.js**, enhancing usability and reducing patient scheduling errors by implementing real-time input validation, dynamic form feedback, and modular component design

SOFTWARE ENGINEERING PROJECTS

Thread-Safe Analytics Server – Scalable RPC server for CSV and Parquet aggregation | Python, gRPC, PyArrow

- Designed and implemented a **multithreaded analytics server** to handle high-volume CSV uploads via **gRPC**, enabling concurrent file processing and aggregation with **per-request RPC isolation**
- Converted incoming CSVs to **Parquet** and persisted them to disk, supporting **low-memory** operation and incremental table growth across hundreds of files
- Optimized columnar query performance by leveraging **PyArrow** to selectively read target columns during RPC requests, achieving up to **60x faster** reads compared to naive full-table loads
- Ensured thread safety with **fine-grained locking** for shared metadata access while minimizing lock contention during disk I/O, increasing concurrent request throughput without race conditions

Loan Analytics with Spark – Scalable loan approval prediction pipeline | PySpark, Hive, HDFS, Docker, SQL

- Built a containerized **Spark cluster** using **Docker Compose** with **NameNode**, **DataNode**, **Spark master**, and **workers** to enable distributed data processing in a local cloud-like environment
- Processed and queried large-scale loan data using **PySpark's RDD**, **DataFrame**, and **SQL** interfaces, joining relational datasets from **Hive** to extract meaningful insights

HDFS Fault-Tolerant Analytics – Distributed data storage and retrieval | Docker, HDFS, PyArrow, WebHDFS

- Deployed a **containerized HDFS cluster** with **NameNode** and multi-**DataNode** setup using **Docker Compose**, enabling distributed block storage and replication of large Parquet files
- Developed a **PyArrow pipeline** to **read and aggregate columnar data** directly from HDFS, optimizing I/O performance by selecting only required columns and achieving up to **50x speedup** compared to full-table reads
- Simulated node failure scenarios to measure **data loss resilience**, verifying block recovery behavior via **WebHDFS** and adapting application logic to minimize impact of partial failures

TECHNICAL SKILLS AND CERTIFICATIONS

- Languages:** Go, Python, C, C++, JavaScript, PHP, SQL, Bash
- Technologies:** Vue.js, CodeIgniter 4, MySQL, Docker, GCP, AWS, Apache Kafka, Apache Spark, Apache Hadoop, Apache Cassandra, gRPC, WebSockets, FUSE, Cron, Git, REST APIs, CI/CD, Unit Testing, Serverless Computing, Distributed Systems